

$$S[\cdot] = \{1, 1, 0, 0, 1, 1, 0, 0\}$$

$$h[\cdot] = \left\{ \frac{1-\sqrt{7}}{16\sqrt{2}}, \frac{5+\sqrt{7}}{16\sqrt{2}}, \frac{14+2\sqrt{7}}{16\sqrt{2}}, \frac{14-2\sqrt{7}}{16\sqrt{2}}, \frac{1-\sqrt{7}}{16\sqrt{2}}, \right. \\ \left. \frac{-3+\sqrt{7}}{16\sqrt{2}} \right\}$$

$$g[\cdot] = \left\{ \frac{-3+\sqrt{7}}{16\sqrt{2}}, \frac{-1+\sqrt{7}}{16\sqrt{2}}, \frac{14-2\sqrt{7}}{16\sqrt{2}}, \frac{-14-2\sqrt{7}}{16\sqrt{2}}, \right. \\ \left. \frac{5+\sqrt{7}}{16\sqrt{2}}, \frac{-1+\sqrt{7}}{16\sqrt{2}} \right\}$$

$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	0	0	1
$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0	0	1
0	0	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	0
0	0	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0
$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	1
$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	1
$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	0
$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0
$\frac{\sqrt{2}}{8}$	$\frac{\sqrt{14}}{8}$	$\frac{7\sqrt{2}}{8}$	$-\frac{\sqrt{14}}{8}$	$\frac{\sqrt{2}}{8}$	$\frac{\sqrt{14}}{8}$	$\frac{7\sqrt{2}}{8}$	$-\frac{\sqrt{14}}{8}$	

$$\{1, 1, 0, 0, 1, 1, 0, 0\}$$

4EJ ↓

8EJ ↓

$$\left\{ \frac{\sqrt{2}}{8}, \frac{7\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, \frac{7\sqrt{2}}{8} \right\}$$

$$\left\{ \frac{\sqrt{14}}{8}, -\frac{\sqrt{14}}{8}, \frac{\sqrt{14}}{8}, -\frac{\sqrt{14}}{8} \right\}$$

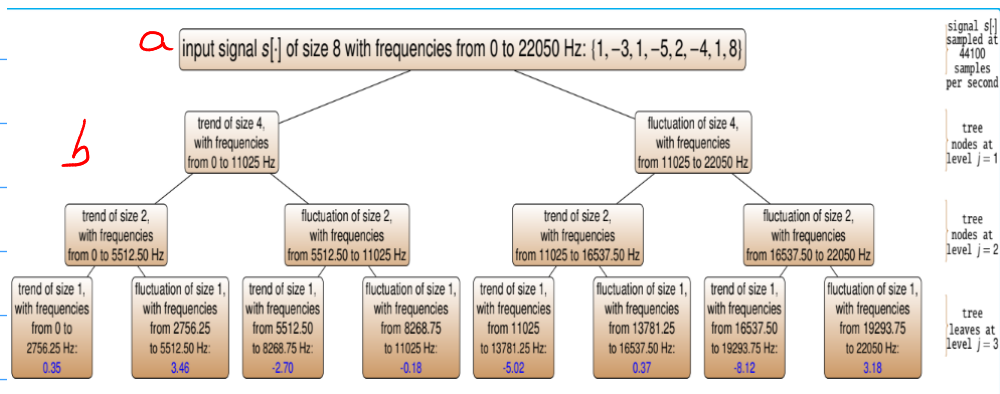
$$DTWPT, (SEJ) = \left\{ \frac{\sqrt{2}}{8}, \frac{7\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, \frac{7\sqrt{2}}{8}, \frac{\sqrt{14}}{8}, \right. \\ \left. -\frac{\sqrt{14}}{8}, \frac{\sqrt{14}}{8}, -\frac{\sqrt{14}}{8} \right\}$$

$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{\sqrt{2}}{8}$
$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{7\sqrt{2}}{8}$
0	0	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{\sqrt{2}}{8}$
0	0	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{7\sqrt{2}}{8}$
$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{\sqrt{2}}{8}$
$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{7\sqrt{2}}{8}$
$\frac{14+2\sqrt{7}}{16\sqrt{2}}$	$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{1-\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{\sqrt{2}}{8}$
$\frac{14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{-14-2\sqrt{7}}{16\sqrt{2}}$	$\frac{5+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	0	0	$\frac{-3+\sqrt{7}}{16\sqrt{2}}$	$\frac{-1+\sqrt{7}}{16\sqrt{2}}$	$\frac{7\sqrt{2}}{8}$

1,

$$\frac{26}{06} = 26$$

$$X = 44100 \text{ Hz} / 2$$



$$a \rightarrow \frac{X}{2}$$

$$b_1 \rightarrow \frac{\frac{X}{2}}{\frac{L}{1}} \rightarrow \frac{X}{4}$$

$$b_2 \rightarrow \frac{X}{4} \rightarrow \frac{X}{2}$$

$$2^k \cdot \frac{X}{2 \cdot 2^k}$$

$$k=0, y=0 \rightarrow 1 \cdot \frac{X}{2 \cdot 1} = \frac{X}{2}$$

$$k=0, y=1 \rightarrow 1 \cdot \frac{X}{2 \cdot 2} = \frac{X}{4}$$

$$k=1, y=1 \rightarrow 2 \cdot \frac{X}{2 \cdot 2} = \frac{X}{2}$$

$$k=0, y=2 \rightarrow 1 \cdot \frac{X}{2 \cdot 2^2} = \frac{X}{8}$$

$$k=1, y=2 \rightarrow 2 \cdot \frac{X}{2 \cdot 2^2} = \frac{X}{4}$$

$$k=2, y=2 \rightarrow 2^2 \cdot \frac{X}{2 \cdot 2^2} = \frac{X}{2}$$

input signal $s[\cdot]$ of size 8 with frequencies from 0 to 22050 Hz: $\{1, -3, 1, -5, 2, -4, 1, 8\}$

tree
leaves at
level $j = 3$

$$(k+1) \frac{1}{2^{(k+1)}} \times$$

trend of size 4,
with frequencies
from 0 to 11025 Hz

fluctuation of size 4,
with frequencies
from 11025 to 22050 Hz

trere
with
from 0

trend of size 2,
with frequencies
from 0 to 5512.50 Hz

fluctuation of size 2,
with frequencies
from 5512.50 to 11025 Hz

trend of size 2,
with frequencies
from 11025 to 16537.50 Hz

fluctuation of size 2,
with frequencies
from 16537.50 to 22050 Hz

trend of size 1,
with frequencies
from 0 to
2756.25 Hz:
0.35

fluctuation of size 1,
with frequencies
from 2756.25
to 5512.50 Hz:
3.46

trend of size 1,
with frequencies
from 5512.50
to 8268.75 Hz:
-2.70

fluctuation of size 1,
with frequencies
from 8268.75
to 11025 Hz:
-0.18

trend of size 1,
with frequencies
from 11025
to 13781.25 Hz:
-5.02

fluctuation of size 1
with frequencies
from 13781.25
to 16537.50 Hz:
0.37

trend of size 1,
with frequencies
from 16537.50
to 19293.75 Hz:
-8.12

fluctuation of size
with frequency
from 19293.75
to 22050 Hz:
3.18

$$b \rightarrow \min_k \frac{X}{2^{(k-1)}}$$

b	→	1	1	1	$2' = 2$
		0	X	X	
			1		

c $\rightarrow$ $\begin{matrix} 1 & 1 & 1 & 1 & 1 \\ 0 & \frac{1}{8} & \frac{x}{4} & \frac{2x}{8} & x_2 \end{matrix}$ $2^4 = 16$

$$\frac{x}{7} + \frac{x}{8} = \frac{8x + 7x}{56} = \frac{15x}{56} = \frac{3x}{11.2}$$

$$(k+1) \frac{1}{2^{(k+1)}} x$$

input signal $s[\cdot]$ of size 8 with frequencies from 0 to 22050 Hz: {1, -3, 1, -5, 2, -4, 1, 8}

signal $s[\cdot]$
sampled at
44100
samples
per second

tree
nodes at
level $j=1$

tree
nodes at
level $j=2$

tree
leaves at
level $j=3$

$$(k+1) \frac{1}{2^{(k+1)}} \quad X$$

$$k=0; j=0 \rightarrow \frac{1 \cdot X}{2^1} = \frac{X}{2}$$

$$k=0; j=1 \rightarrow \frac{1 \cdot X}{2^2} = \frac{X}{4} \quad / \quad k=1; j=1 \rightarrow \frac{2 \cdot X}{2^2} = \frac{X}{2}$$

$$k=0; j=2 \rightarrow \frac{1 \cdot X}{2^3} = \frac{X}{8} \quad / \quad k=1; j=2 \rightarrow \frac{2 \cdot X}{2^3} = \frac{X}{4} = \frac{X}{4}$$

$$k=2; j=2 \rightarrow \frac{3 \cdot X}{2^3} = \frac{3X}{8} \quad / \quad k=3; j=2 \rightarrow \frac{4 \cdot X}{2^3} = \frac{X}{2}$$

$$k=0; j=3 \rightarrow \frac{1 \cdot X}{2^4} = \frac{X}{16}$$

$$k=1; j=3 \rightarrow \frac{2 \cdot X}{2^4} = \frac{X}{8} = \frac{X}{8}$$

$$k=2; j=3 \rightarrow \frac{3 \cdot X}{2^4} = \frac{3X}{16}$$

$$k=3; j=3 \rightarrow \frac{4 \cdot X}{2^4} = \frac{X}{4} = \frac{X}{4}$$

$$k=4; j=3 \rightarrow \frac{5 \cdot X}{2^4} = \frac{5X}{16}$$

$$k=5; j=3 \rightarrow \frac{6 \cdot X}{2^4} = \frac{6X}{16}$$

$$k=6; j=3 \rightarrow \frac{7 \cdot X}{2^4} = \frac{7X}{16}$$

$$k=7; j=3 \rightarrow \frac{8 \cdot X}{2^4} = \frac{X}{2}$$